Visual Search Experiment
PSY310 Lab in Psychology
Lab Report

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GitHub
Link:https://github.com/tamannaz06/Tamanna-Chandna_VisualSearch

Introduction

Visual search tasks are commonly used for studying attention as it requires an observer to focus on the task of finding the target and while doing so, ignore the distractors which are simultaneously present.

“In a typical visual search task, an observer is asked to determine whether a target stimulus is present among a variable number of distractor stimuli. The total number of stimuli is here referred to as the display set size and has often been shown to affect performance” (Palmer et al., 1993).

Early studies have found that search is easy when the target differs from all its distractors along any particular feature dimension. This could be color, shape or even orientation. This is called parallel search. However, when it’s a conjunction search, focal attention is needed to conjoin visual features into single object files. This is called serial search as Reaction time increases proportionately depending on the number of distractors because some of the distractors also contain target features and the target cannot be detected on the basis of individual feature maps. (Chan & Hayward, 2013). Slopes can be made for the same with Reaction time on the Y axis and number of distractors or set size on the X axis. The analysis of RT × size slopes has been widely used to propose different theories on how the brain works for such tasks (RODRIGUEZ-SANCHEZ et al., 2007). Rodrigues-Sanchez et al (2007) also stated that flat slopes indicate that attention was deployed efficiently while steep slopes show that attention got narrowed in certain areas and hence may take longer to finish the task.

Hence, through performance measures like accuracy and reaction time taken to find the target in visual search studies we can study how efficiently attention is allocated.

Additionally, according to Lavie and Cox (1997), sometimes despite efficient search, selective attention being allocated equally is not guaranteed. This is due to their Load Theory of attention which states that when search tasks are easy, participants' capacity is more easily influenced by non-target stimuli and thus create greater distraction as compared to high cognitive load situations as then there are no additional resources available for attending to task irrelevant stimuli (Lavie & Cox, 1997).

Method

Participants

This experiment has five participants all of whom are undergraduate college students of Ahmedabad University between the ages 18-22. There are both male and female participants. The aim was to assess the time taken by them to find the target (T) as well as the accuracy of responses.

Assigned no.	Age	Sex	Dominant hand	Year of study
#1	19	Male	Right	3rd
#2	22	Male	Right	4th
#3	18	Female	Right	1st
#4	21	Female	Right	3rd
#5	19	Female	Right	2nd

Table 1: Details of the participants

Materials

The experiment was conducted on a laptop with PsychoPy Studio 2026.2.4 installed. The laptop has a monitor of dimension 13.3”, a resolution of 1920x1080 pixels and a refresh rate of 60 Hz. 200 trials of the experiment were conducted.
The stimulus was a

Stimuli

Each trial had a target which was a capital letter ‘T’ and either five or ten distractors which were represented by capital letter ‘L’. These distractors had two characteristics controlled. 1) They rotated randomly between 1° and 360° and 2) They had either 5 or 10 distractors across the 200 trials appear in a random order, making a 50% split of each condition. Hence, participants completed a total of 200 trials each - 100 with set-size 5 and 100 with set-size 10, allowing a consideration of visual load. All the characters were white and presented on a grey background with a height of 0.1.

On each of the trials, the target and distractors (five or ten) were randomly distributed across the screen, such that the target was positioned in different locations each time. Due to this spatial spread, on some trials half of the letters were obscured as if they had been hidden beyond or behind the screen’s limits. However, in such cases too, the target was clear and easily detectable.

Figure 1 and Figure 2 are screenshots from the trials in the two conditions.

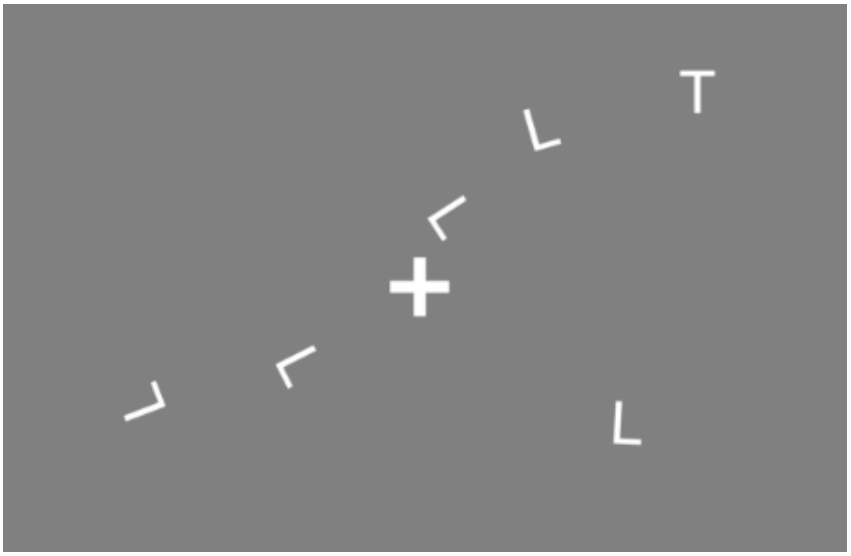


Figure 1: Search array with distractor set size=5

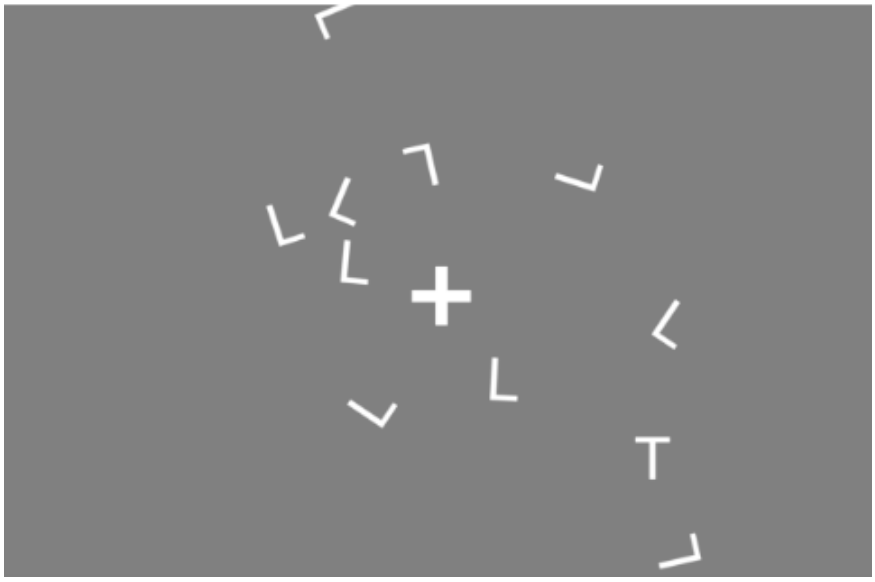


Figure 2: Search array with distractor set size=10

Procedure

Each trial began with a cross-fixation and the visual search array. This remained on the screen until the participant clicked on the target, that is valid click: if the participant made a mistake and clicked on anything other than the target, the trial would continue until the target is found.
To note: The initial 2 datasets have ‘any click’ as that’s what the tutorial video had specified.

Participants were given instructions to locate the target ‘T’ as quickly as possible and click on the target with the help of a mouse. Once a valid mouse click on the target had been made, the trial ended. Figure 3 shows the accuracy of each participant. The search is said to be accurate only if the target was found on first click.

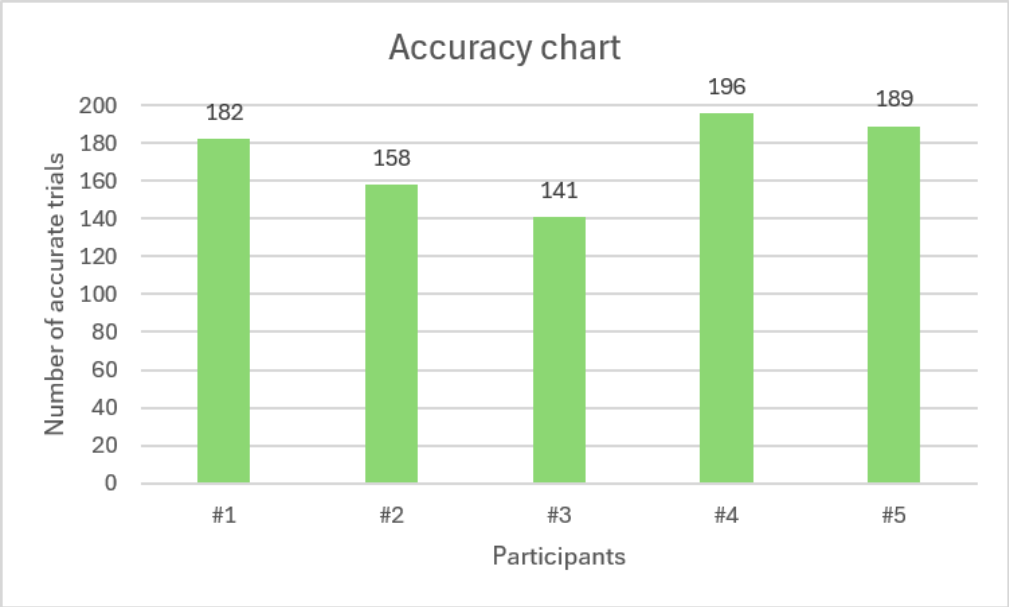


Figure 3: Number of accurate trials out of 200 trials per participant

Results

The mean reaction times (RTs) for only the accurate trials were calculated separately for set size 5 and set size 10 for each participant. The slope or visual search efficiency measure was calculated by dividing the change in RT by the change in set size.

No. of Accurate Trials	Mean RT for all accurate trials (s)	Mean RT for 5 Distractors (s)	Mean RT for 10 distractors (s)	Slope
173.2	1.700	1.503936088	1.9088267475	0.080978

Table 2: This shows the total averages obtained from all the 5 participants cumulatively of accurate trials, average RTs across set sizes 5 and 10, and slope/search efficiency.

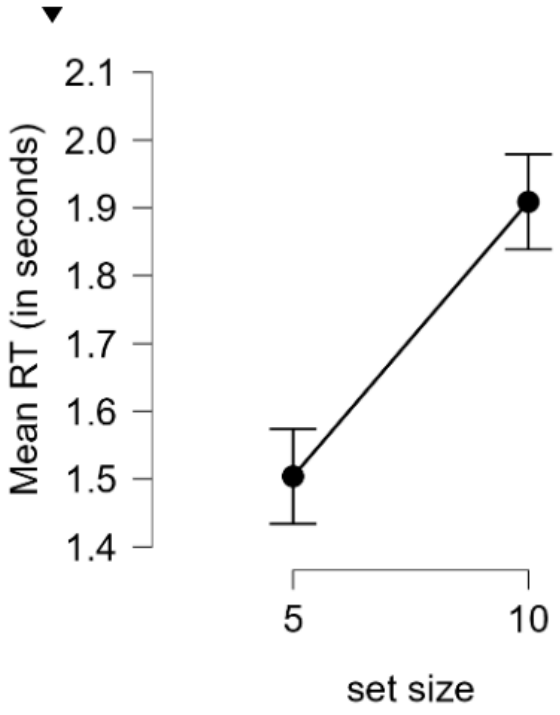


Figure 4: Mean RTs of all 5 participants for set size 5 and 10 (generated on JASP)

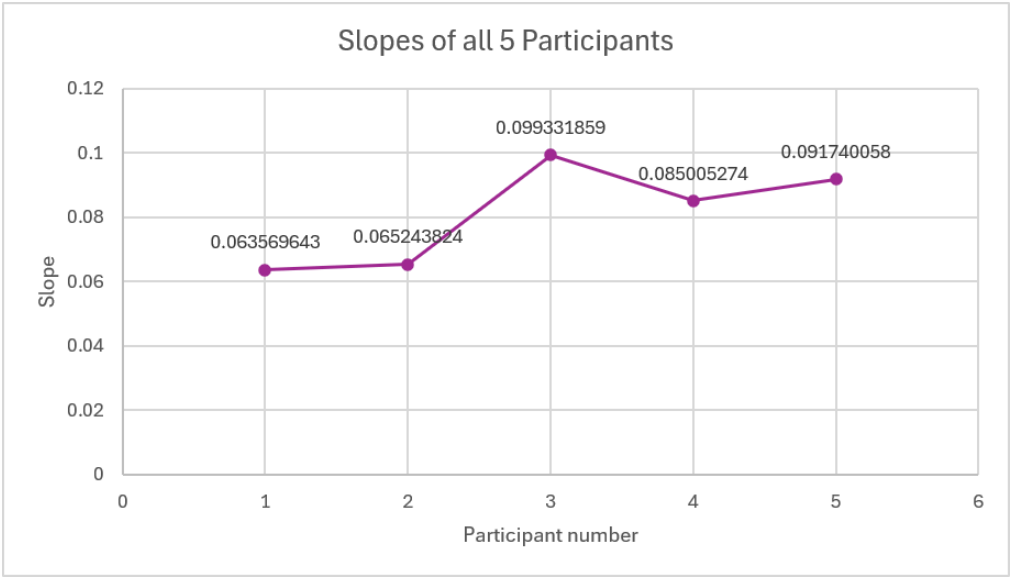
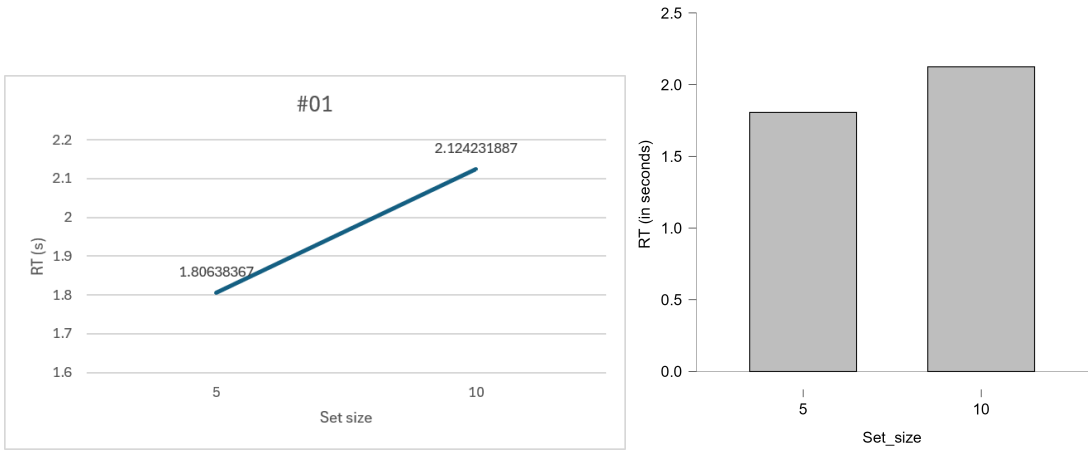


Figure 5: Slopes of all participants performance calculated with RT and Set size

The RTs for set size 5 and 10 show a statistically significant difference with participants taking longer to find the target in the presence of 10 distractors. Each participant's data is individually analyzed below.

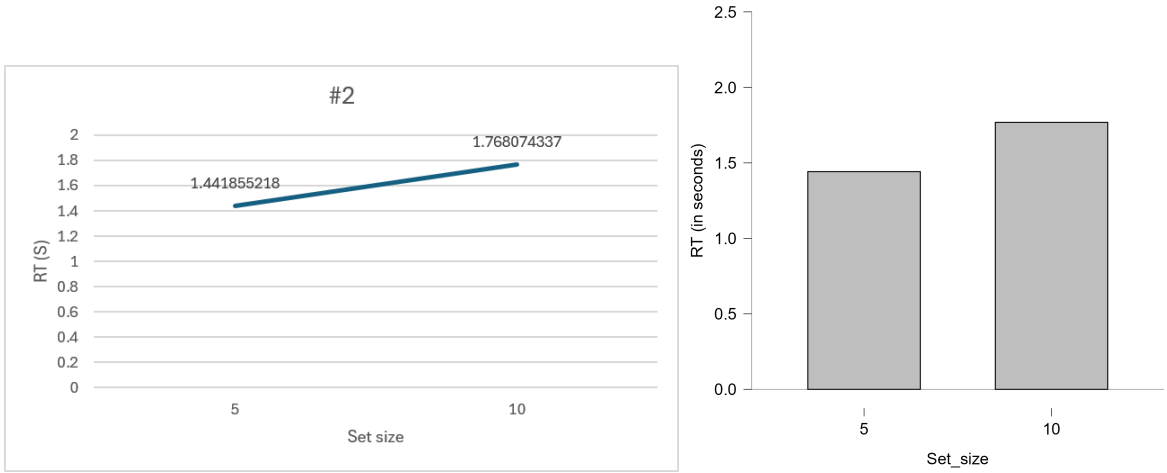
Participant 1



A line plot and bar plot with the average reaction time versus the set size of 5 and 10 for participant 1

The mean RT for both set sizes combined was 1.965 seconds with an increase of 0.31 seconds when moving from 5 to 10 distractors. The slope (0.06357 s/item) indicates that for every additional item added to the visual display, it takes the participant an average of 63.57 milliseconds longer to find the target.

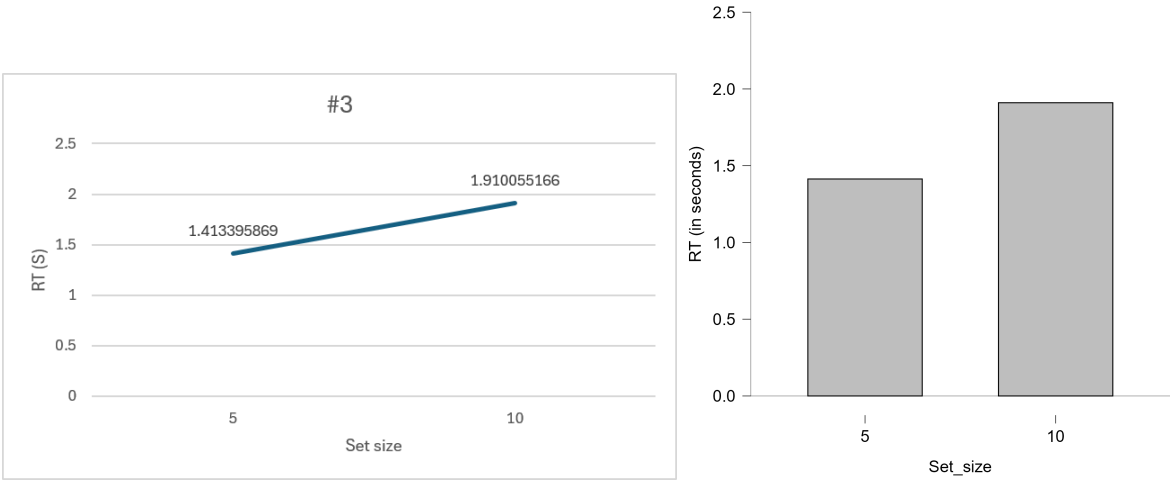
Participant 2



A line plot and bar plot with the average reaction time versus the set size of 5 and 10 for participant 2

The mean RT for both set sizes combined was 1.607 seconds with an increase of 0.3262 seconds when moving from 5 to 10 distractors. The slope (0.065344 s/item) indicates that for every additional item added to the visual display, it takes the participant an average of 65.344 milliseconds longer to find the target.

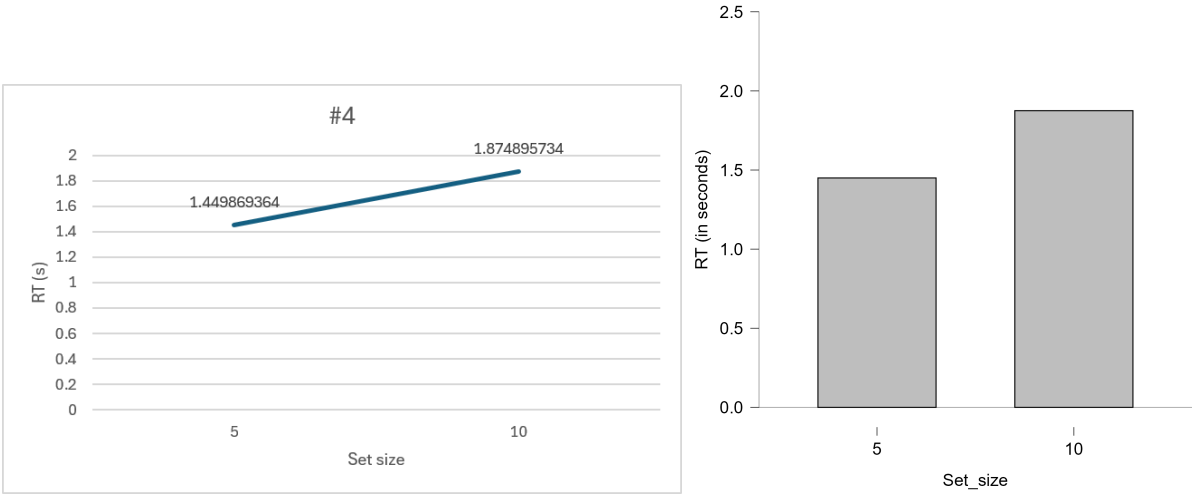
Participant 3



A line plot and bar plot with the average reaction time versus the set size of 5 and 10 for participant 3

The mean RT for both set sizes combined was 1.6705 seconds with an increase of 0.496 seconds when moving from 5 to 10 distractors. The slope (0.099 s/item) indicates that for every additional item added to the visual display, it takes the participant 99 milliseconds longer to find the target.

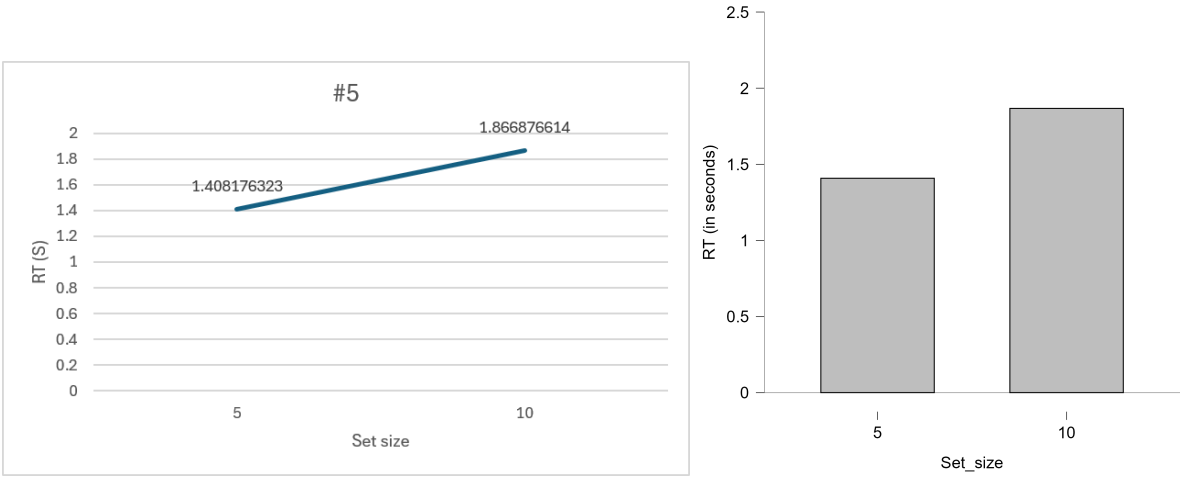
Participant 4



A line plot and bar plot with the average reaction time versus the set size of 5 and 10 for participant 4

The mean RT for both set sizes combined was 1.6558 seconds with an increase of 0.425 seconds when moving from 5 to 10 distractors. The slope (0.085 s/item) indicates that for every additional item added to the visual display, it takes the participant an average of 85 milliseconds longer to find the target.

Participant 5



A line plot and bar plot with the average reaction time versus the set size of 5 and 10 for participant 5

The mean RT for both set sizes combined was 1.602 seconds with an increase of 0.4587 seconds when moving from 5 to 10 distractors. The slope (0.09 s/item) indicates that for every additional

item added to the visual display, it takes the participant an average of 90 milliseconds longer to find the target.

Discussion

The steepness of the RT by set size function is generally taken as an indicator of the efficiency with which attention is allocated (Treisman & Gelade, 1980). It shows how much each extra item in the display adds to search tim. A slope close to 0 indicates the target is detected near the same rate for all number of distractors, implying that it is attention grabbing. If the slope is positive, then objects are being inspected one at a time or in small groups, in which case each additional distractor increases the search time. This was the case in all of the participants' data above.

This current experiment indicated clearly positive slopes for all five participants ranging from 0.064 s/item (Participant 1) to 0.099 s/item (Participant 3). The mean slope was found to be approximately 81ms per item (or 0.081 s). With 10 distractors, Mean RT on correct trials was increased to 1.91 s, and this increase was seen in all participants compared to 5 distractors which was 1.50 s. This suggests ineffective search as attention was needed to be dragged across the display rather than drawn directly to the target. It confirms Duncan and Humphreys (1989) theorizing that the harder the search, the more items are competing for attention and so each item must be assessed and rejected before the desired item can be chosen.

The direction of the effect was uniform but the size or amount was variable. The lowest slopes were for Participant 1 (0.064) and Participant 2 (0.065), the steepest was for Participant 3 (0.099), and Participants 4 and 5 were in the middle (0.085 and 0.092, respectively). Participant 1 was also the slowest overall (1.81 s at set size 5), meaning that a slow baseline did not necessarily result in a steep slope. These differences may be due to differing search tactics such as searching thoroughly versus rapidly.

Certain restrictions need to be kept in mind. Since only two set sizes were used, there is no way to determine if growth is linear with set size for RT. Only accurate trials were used and the sample was pretty small (n=5). Some noise in RTs could be introduced in case of confounds like location of experiment or due to the individual participant difference variables like amount of sleep. They will not be able to account for an increase that is observed in all five participants. In general, the positive slopes indicate the decrease of search efficiency as the number of elements in the set grew.

References

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